GRUPA 10

25.10.2026.

1. SKOKOVNI ODZIV SUSTAVA DRUGOG REDA

$$G_1(s) = \frac{K_1}{s^2 + a_1 s + a_0}$$

PODACI: $K_1 = 27$, $a_1 = 4$, $a_0 = 36$

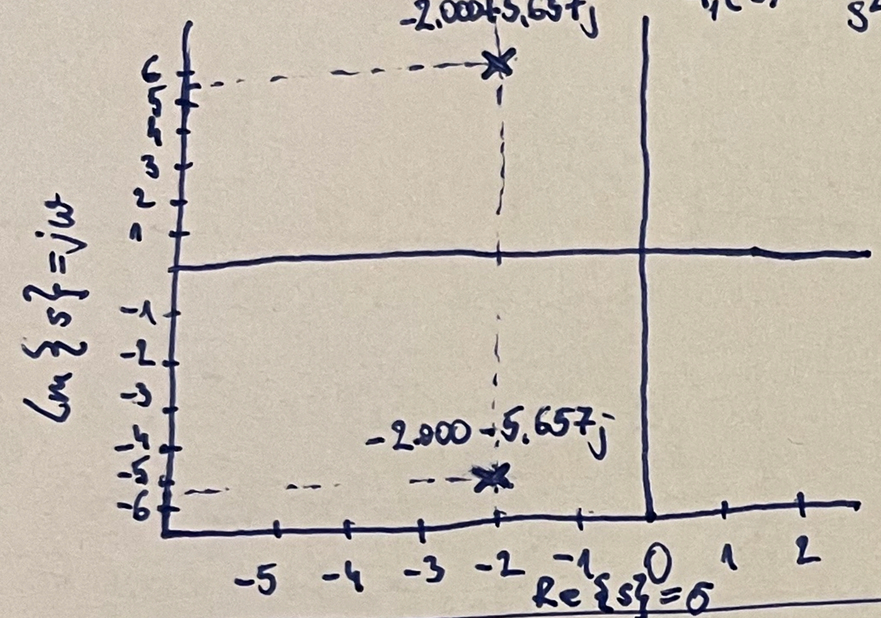
NA ULAZ SUSTAVA DJELUJE JEDINIČNA STEPENICA $S(t)$

a) POLOVI SUSTAVA I PRIKAZ U KOMPLEKSNOJ S-RAVNINI

$$s^2 + 4s + 36 = 0$$

$$s_{1,2} = \frac{-4 \pm \sqrt{16 - 144}}{2} = -2 \pm j\sqrt{32} = -2.000 \pm j5.657$$

$$G_1(s) = \frac{27}{s^2 + 4s + 36}$$



b) FREKVENCIJA OSCILACIJA ω_n I FAKTOR PRIGUŠENJA ζ USPOREDBOM S OPĆIM OBLIKOM

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

$$\omega_n^2 = 36$$

$$\omega_n = \sqrt{36} = 6 \text{ RAD/s}$$

$$\boxed{\omega_n = 6{,}000 \text{ RAD/s}}$$

$$2 \cdot \zeta \cdot \omega_n = 4$$

$$2 \cdot \zeta \cdot 6 = 4 \;/:12$$

$$\zeta = \frac{1}{3}$$

$$\boxed{\zeta = 0{,}333}$$

c) VRIJEME PRVOG MAKSIMUMA T_p, POSTOTAK NADVIŠENJA P.O. I VRIJEME SMIRIVANJA T_s.

$$\omega_d = \omega_n \cdot \sqrt{1 - \zeta^2} = 6 \cdot \sqrt{1 - 0.333^2}$$

$$\omega_d = 5.657 \text{ RAD/s}$$

$$\boxed{T_p = \frac{\pi}{\omega_d} = \frac{\pi}{5.657} = 0.555 \text{ s}}$$

$$P.O. = 100 \cdot e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} = 100 \cdot e^{-\frac{0.333\pi}{0.943}}$$

$$P.O. = 100 \cdot e^{-\frac{1.047}{0.943}} = 100 \cdot e^{-1.110}$$

$$\boxed{P.O. = 32{,}956\ \%}$$

$$\boxed{T_s = \frac{4}{\zeta \cdot \omega_n} = \frac{4}{0.333 \cdot 6} = 2{,}000 \text{ s}}$$

d) ANALITIČKI IZRAZ SKOKOVNOG ODZIVA $y(t)$ RASTAVOM NA PARCIJALNE RAZLOMKE I LAPLACEOVIM PAROVIMA ZA $e^{-\alpha t}\sin(\omega t)$ i $e^{-\alpha t}\cos(\omega t)$.

$\boxed{s(t)=1}$ KOD JEDINIČNE STEPENICE ZA $t>0$

ODZIV ZA JED. STEPENICU: $y(t)=\mathcal{L}^{-1}\{G(s)\frac{1}{s}\}$

$$y(t)=\mathcal{L}^{-1}\left\{\frac{27}{s^2+4s+36}\cdot\frac{1}{s}\right\}=\mathcal{L}^{-1}\left\{\frac{27}{s(s^2+4s+36)}\right\}$$

$$\frac{27}{s(s^2+4s+36)}=\boxed{\frac{A}{s}+\frac{Bs+C}{s^2+4s+36}}$$

$$27=A(s^2+4s+36)+s(Bs+C)=(A+B)s^2+(4A+C)s+36A$$

s^0: $36A=27$ $\quad$ s^1: $4A+C=0$ $\quad$ s^2: $A+B=0$

$A=\frac{3}{4}=0.75$ $\quad$ $4\cdot\frac{3}{4}+C=0$ $\quad$ $0.75+B=0$

$\boxed{A=0.75}=\frac{3}{4}$ $\quad$ $\boxed{C=-3}$ $\quad$ $\boxed{B=-0.75}=-\frac{3}{4}$

$$y(t)=\mathcal{L}^{-1}\left\{\frac{0.75}{s}-0.75\,\frac{s+4}{s^2+4s+36}\right\}$$

$s+4=(s+2)+2$

$s^2+4s+36=(s+2)^2+32$
$=(s+2)^2+(4\sqrt{2})^2$
$=(s+2)^2+5.657^2$

$$=\mathcal{L}^{-1}\left\{\frac{0.75}{s}-0.75\cdot\frac{(s+2)+2}{(s+2)^2+5.657^2}\right\}$$

$$=\mathcal{L}^{-1}\left\{\frac{0.75}{s}-0.75\cdot\frac{s+2}{(s+2)^2+5.657^2}-0.265\cdot\frac{5.657}{(s+2)^2+5.657^2}\right\}$$

$-0.75\cdot\frac{(s+2)+2}{(s+2)^2+5.657^2}$

$$=0.75\,s(t)-0.75\cdot e^{-2t}\cos(5.657t)\cdot s(t)-0.265\cdot e^{-2t}\sin(5.657t)\cdot s(t)$$

$$=-0.75\cdot\frac{s+2}{(s+2)^2+5.657^2}-0.75\cdot\frac{2}{(s+2)^2+5.657^2}$$

$$\frac{1.5}{(s+2)^2+5.657^2}\cdot\frac{5.657}{5.657}$$

$$=\frac{1.5}{5.657}\cdot\frac{5.657}{(s+2)^2+5.657^2}$$

$$=0.265\cdot\frac{5.657}{(s+2)^2+5.657^2}$$

$$\boxed{y(t)=0.750-0.750\cdot e^{-2t}\cos(5.657t)-0.265\cdot e^{-2t}\sin(5.657t)}$$

e) VRIJEDNOST ODZIVA U USTALJENOM STANJU $y(\infty)$ I GREŠKA e_{ss}.

$$y(\infty)=\lim_{s\to 0}\ s\cdot G(s)\cdot\mathcal{L}\{s(t)\}$$

$$y(\infty)=\lim_{s\to 0}\ s\cdot\frac{27}{s^2+4s+36}\cdot\frac{1}{s}$$

$$=\lim_{s\to 0}\ \frac{27}{0^2+4\cdot 0+36}=\frac{27}{36}=0.750$$

$$\boxed{y(\infty)=0.750}$$

$$e_{ss}=1-y(\infty)=1-0.750=0.250$$

$$\boxed{e_{ss}=0.250}$$